



Predict With Events

Quick facilitation notes & answer key

Goal

Students PREDICT the combined outcome of an event chain before running it — the C3.2 'predict what happens when both start' skill applied to events. They must follow green flag → Bit moves + broadcasts 'go' → Pixel runs on the message, then state both landings. Predicting the whole chain (not just one script) is the target.

Ontario curriculum (Grade 2 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: students write their prediction on a sticky note before tracing the events with two counters on a 0-5 path.

Run it (about 20 min)

1. I DO: model predicting Bit — 'green flag, 1 forward 3 → 4, then broadcast go. I predict Bit on 4.'
2. WE DO: predict Pixel — 'it receives go, 5 back 2 → 3. I predict Pixel on 3.' Commit predictions before running.
3. YOU DO: students run the chain, compare to their prediction, and name the event that started Pixel (the message 'go').
4. Connect: 'A good prediction follows every event, not just the first one.'

Answer key (modelled by the run-gate)

Ex 1 (predict) — Bit: 1 → forward 3 → 4 (predict 4). Pixel: 5 → back 2 → 3 (predict 3).

Ex 2 (run) — the run confirms Bit on 4 and Pixel on 3; matching predictions.

Ex 3 — Pixel was started by the broadcast message 'go' (its 'when I receive go' hat).

Chain note: the green flag only directly starts Bit; Pixel's start depends on Bit's broadcast, so the prediction must include the message step.

Watch for & differentiate

Common error: predicting only Bit — the prediction must cover both, including the message-triggered Pixel.

Common error: thinking Pixel starts on the green flag — it starts on the 'go' message Bit sends.

Simplify: predict Bit fully and run it, then predict Pixel and run it, before combining.

Extend: ask what Pixel would do if Bit's broadcast block were removed (Pixel never starts, stays on 5).

Success indicator: the child predicts Bit 4 and Pixel 3, confirms by running, and names the 'go' message as Pixel's trigger.